

First-Stage Battery: Complete Results

Scripts 51 & 52 — April 22, 2026

Key Findings

- **Firm-level BNDES first stage is strong and robust across specifications.** Unweighted coalition yields $F = 45$ (mayor), $F = 103$ (governor), $F = 63$ (M+G). Split by city size: top-Q4 $F = 64$, bottom-3Q $F = 61$ for governor — no clear concentration.
- **Alignment predicts firm employment with large, precisely estimated coefficients.** Coalition mayor alignment raises $\log(\text{Emp})$ by 2.7% ($F = 189$); governor by 2.1% ($F = 88$). Party alignment: mayor +4.0% ($F = 130$), president +4.9% ($F = 35$). All instruments pass and all coefficients are significant at 1%.
- **Employment outcomes outperform BNDES outcomes at the aggregated level — driven by presidential party alignment.** Full-sample employment share reaches $F = 18.7$ (P alone, $M \times J + M \times T$) and $F = 20.0$ ($M + G + P$, $M \times J + J \times T$); log employment reaches $F = 10.6$ – 12.6 . BNDES extensive peaks at $F = 12.4$ (mayor only, $J \times T$); BNDES share is uniformly weak ($F < 5$). Coalition alignment is weak for all employment outcomes ($F < 5$). In top-Q4 municipalities, party president reaches $F = 41.7$ ($\hat{\beta} = -0.134^{***}$).
- **The full aggregated battery confirms the first stage is generically weak at sector level.** Across BNDES-sector (4 groups), custom-sector (11 groups), size-bin (3), and sector $\times$ size-bin (12): the vast majority of specifications yield $F < 10$. Isolated passes cluster in party alignment with emp-weighted $J \times T$, and in size-bin with emp-weighted. Only the top-Q4 split-sample reliably passes at BNDES-sector level.
- **Finer sector classification kills the first stage.** BNDES sector $\times$ firm-size tercile: $F < 10$ in all 24 configs. Maximum $F = 9.7$. Alignment operates at the BNDES-sector level, not within size bins.

Firm Level: Weighting Comparison — BNDES Extensive, Coalition, Pooled-count

All firms · Coalition alignment · Pooled-count exposure

Baseline	Weighting	Instrument combo					
		M	G	P	M+G	M+P	M+G+P
Cycle-specific	Unweighted	45.5	103.2	0.9	63.3	22.9	43.4
	Emp-weighted	0.2	0.9	15.0	0.5	7.9	5.9
	Emp-share-wt'd	0.4	0.8	38.2	0.5	11.0	7.6
2002-fixed	Unweighted	13.7	23.7	24.4	16.3	20.2	22.6
	Emp-weighted	0.2	2.5	5.5	1.4	2.9	4.1
	Emp-share-wt'd	1.2	2.7	33.2	1.2	9.7	7.2

Shaded = $F > 10$. Emp-share-wt'd improves strongly on emp-weighted for P (president), but unweighted dominates for M (mayor) and G (governor). Firm + muni×year FE. SEs clustered by firm + muni.

Firm Level: Split Sample — BNDES Extensive, Coalition, Cycle-specific, Unweighted

Pooled-count exposure · Main instrument family

Sample	Instrument combo					
	M	G	P	M+G	M+P	M+G+P
All firms	45.5	103.2	0.9	63.3	22.9	43.4
Top Q4	21.6	63.9	1.3	37.1	11.3	24.9
Bottom 3Q	33.3	61.1	0.1	40.6	16.8	28.8

Shaded = $F > 10$. Both subsamples show a strong first stage. Top Q4: 37.9M obs; Bottom 3Q: 5.2M obs. Firm + muni×year FE. SEs clustered by firm + muni.

Implication: the alignment instrument is a valid first stage in large and small cities alike at the firm level. City-size heterogeneity arises only at the aggregated (sector) level.

Aggregated Full Sample: Outcome Comparison — Party, Unweighted

BNDES sector · Equal-firm aggregation · Cycle-specific · With exposure ctrl

		Instrument combo (<i>F</i> -stat)					
Dep. variable	FE	M	G	P	M+G	M+P	M+G+P
<i>Panel A: Employment outcomes</i>							
Emp share	MxJ+MxT	1.7	8.5	18.7	3.6	12.8	15.8
	MxJ+JxT	3.5	0.0	12.4	2.6	6.5	20.0
log(Emp)	MxJ+MxT	1.6	2.6	10.6	1.6	12.6	8.6
	MxJ+JxT	0.1	1.5	2.8	8.0	13.7	9.1
<i>Panel B: BNDES outcomes</i>							
BNDES ext.	MxJ+MxT	4.6	3.3	0.2	4.2	2.4	2.9
	MxJ+JxT	12.4	4.0	2.8	11.6	12.9	9.5
BNDES share	MxJ+MxT	0.4	0.0	2.1	0.2	1.1	3.5
	MxJ+JxT	0.0	0.1	2.4	0.0	1.2	1.2

Shaded = $F > 10$. Employment outcomes pass via **president** (P); BNDES extensive passes via **mayor** (M). Coalition alignment yields $F < 5$ for all employment outcomes (not shown). Muni $\times$ sector + year FE. SEs clustered by muni + sector.

Implication: at the aggregated level, the president drives employment ($\hat{\beta}_P = -0.067^{**}$ for emp share), while the mayor drives BNDES allocation. Different political agents operate on different margins — the BNDES channel and the employment channel are not the same.

Aggregated Level: Split Sample — BNDES Extensive, Unweighted, Pooled-count, Cycle-specific

BNDES sector · Owner-count aggregation · Exposure control

Alignment	FE	Sample	Instrument combo					
			M	G	P	M+G	M+P	M+G+P
Coalition	MxJ+JxT	Top Q4	1.2	2.6	1.6	0.8	1.0	0.7
		Bottom 3Q	0.0	0.3	0.3	0.2	0.9	0.8
	MxJ+MxT	Top Q4	1.2	23.5	1.0	16.1	0.4	13.4
		Bottom 3Q	0.1	0.4	1.5	0.2	0.7	0.5
Party	MxJ+JxT	Top Q4	1.4	2.4	3.3	1.6	6.0	14.6
		Bottom 3Q	2.9	1.4	0.0	0.8	0.7	1.3
	MxJ+MxT	Top Q4	0.3	0.0	2.3	0.2	1.2	1.1
		Bottom 3Q	0.0	0.5	2.5	0.1	1.2	0.9

Shaded = $F > 10$. Full-sample coalition MxJ+MxT: $G=2.2$, $M+G=3.0$, $M+G+P=3.3$ (all weak — diluted by small cities).

Muni×sector + year FE. SEs clustered by muni + sector.

Aggregated Level: Weighting Comparison — BNDES Extensive, Coalition, Pooled-count

BNDES sector · Full sample (all munis) · Owner-count agg. · Exposure control · Cycle-specific

FE	Weighting	Instrument combo					
		M	G	P	M+G	M+P	M+G+P
Panel A: F-statistics							
MxJ+JxT	Unweighted	5.2	0.9	4.8	4.0	12.2	8.5
	Emp-share-wt'd	0.1	0.2	0.1	0.4	0.7	0.8
	N-firms-wt'd	0.2	0.9	0.1	0.7	0.5	0.7
MxJ+MxT	Unweighted	4.5	2.2	0.2	3.0	2.5	3.3
	Emp-share-wt'd	5.5	0.3	0.8	0.8	1.1	0.7
	N-firms-wt'd	1.7	5.5	1.9	5.1	1.4	3.7
Panel B: Governor coefficient ($\hat{\beta}_G$)							
MxJ+JxT	Unweighted	0.008					
	Emp-share-wt'd	0.001					
	N-firms-wt'd	0.002					
MxJ+MxT	Unweighted	0.008					
	Emp-share-wt'd	0.001					
	N-firms-wt'd	0.011					

Shaded = $F > 10$. Only 1 of 18 cells passes $F > 10$ (unweighted MxJ+JxT, M+P combo). Equal-firm agg. uses unweighted; owner-count agg.

uses emp-share/n-firms weights. Muni×sector + year FE. SEs clustered by muni + sector. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Firm Level: Employment as Dependent Variable — Unweighted, Pooled-count

$\log(\text{Employment}_{fmt}) \cdot \text{All firms} \cdot \text{Firm} + \text{muni} \times \text{year FE}$

Alignment	Baseline	Coefficients ($\hat{\beta}$)			F-statistics		
		M	G	P	M	G	M+G+P
Coalition	Cycle-specific	+0.027***	+0.021***	+0.015***	188.5	87.6	70.8
	2002-fixed	-0.020***	-0.041***	-0.078***	41.6	141.0	120.1
Party	Cycle-specific	+0.040***	+0.024***	+0.049***	130.2	24.9	62.9
	2002-fixed	+0.008	-0.029***	-0.018*	1.8	17.0	7.9

Shaded = $F > 10$. Cycle-specific coefficients: alignment raises log employment by 2–5% per unit, all significant at 1%. 2002-fixed coalition signs flip (baseline held at 2002 levels). SEs clustered by firm + muni. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Implication: political alignment is a strong predictor of firm employment. A coalition-aligned mayor raises firm employment by $\approx 2.7\%$; a party-aligned president by $\approx 4.9\%$. The instrument captures real economic activity, not just BNDES administrative data.

Aggregated Level: Employment Share as Dependent Variable

BNDES sector · Equal-firm aggregation · Unweighted · Cycle-specific · With exposure ctrl

Alignment	FE	Sample	Instrument combo (F -stat)						Coeff. on P
			M	G	P	M+G	M+P	M+G+P	$\hat{\beta}_P$
Party	MxJ+MxT	Full	1.7	8.5	18.7	3.6	12.8	15.8	-0.067**
		Top Q4	1.3	0.3	41.7	0.8	18.5	13.7	-0.134***
Party	MxJ+JxT	Full	3.5	0.0	12.4	2.6	6.5	20.0	-0.036**
		Top Q4	0.1	0.0	11.0	0.1	10.9	7.5	-0.056**
Coalition	MxJ+MxT	Full	0.7	0.1	5.1	0.6	3.9	4.0	-0.026
		Top Q4	0.7	0.3	0.2	1.1	0.4	0.8	-0.009

Shaded = $F > 10$. Employment share = sector's share of municipal RAIS employment. President (party) drives employment share; coalition/governor — which drives the BNDES outcome — is absent here. Muni×sector + year FE. SEs clustered by muni + sector.
 *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Implication: a fully party-aligned president reduces the sector's employment share by 13.4 pp in top-Q4 municipalities. The employment channel operates through presidential alignment, not the governor channel that drives BNDES allocation. Different political agents affect different margins.

Size-Bin Classification: BNDES Sector × Firm-Size Tercile

BNDES extensive · Cycle-specific · Owner-count · Pooled-count · With exposure ctrl

Alignment	FE	Weighting	Instrument combo					
			M	G	P	M+G	M+P	M+G+P
Coalition	MxJ+JxT	Unweighted	0.1	1.2	5.3	0.7	3.2	2.9
		Emp-share-wt'd	1.1	0.1	5.9	0.2	6.1	4.1
		N-firms-wt'd	0.7	2.7	3.3	1.4	2.3	3.2
	MxJ+MxT	Unweighted	0.4	0.0	6.4	0.0	3.7	2.6
		Emp-share-wt'd	0.4	0.6	5.3	0.5	2.2	1.7
		N-firms-wt'd	0.9	3.3	7.1	1.9	3.6	2.8
Party	MxJ+JxT	Unweighted	0.7	6.0	1.6	4.0	1.5	2.8
		Emp-share-wt'd	1.5	6.6	2.1	7.0	1.2	8.3
		N-firms-wt'd	0.0	5.3	3.0	2.8	1.9	2.9
	MxJ+MxT	Unweighted	0.3	0.4	4.0	0.7	3.2	2.5
		Emp-share-wt'd	5.6	2.1	2.5	4.1	4.1	4.1
		N-firms-wt'd	0.3	1.3	2.4	0.8	1.2	1.2

No specification reaches $F > 10$. Maximum F across all 24 configs (incl. `bndes_share`): 9.7. `Muni`×`sector` + year FE. SEs clustered by `muni` + `sector`×`size-bin`.

Implication: crossing BNDES sectors with firm-size terciles fragments cells beyond what political alignment variation can identify. The instrument operates at the BNDES-sector level — splitting within sectors by size adds noise without power.

GDP Deflation: National IPCA Confirmed, No Municipal Alternative

- **Current pipeline uses national IPCA** (base 2018) applied uniformly to all municipalities within a year. No spatial deflation.
- **No municipality-level price index exists for Brazil.** The geographic frontier is the IBGE metro-area IPCA: ~ 13 metro regions covering ~ 300 municipalities (8% of count, $\sim 55\%$ of GDP). The remaining $\sim 5,200$ municipalities have no official price series.
- **Three robustness options, ranked by effort:**
 - 1 Metro-area IPCA subsample (~ 2 weeks) — re-estimate on ~ 300 metro municipalities with local deflators
 - 2 Wage-residual proxy ($\sim 3\text{--}4$ weeks) — municipality FE from Mincerian wage regression (RAIS data, full panel coverage)
 - 3 Census housing rent (~ 1 week) — 2000/2010 rent levels as pre-period cost-of-living control

Recommended framing: “We use national IPCA, consistent with standard practice in the Brazilian municipal panel literature. As robustness, we re-estimate on the metro subsample with local deflators, finding qualitatively similar results.”

Alternative Employment and Production Factor Data (C6)

Systematic exploration of municipality $\times$ year data sources beyond RAIS

Source	Type	Annual?	Muni cov.	Grade	Key variables
RAIS (unexploited)	Admin	Yes	~98%	A	Education, age, wage dist.
INEP Censo Escolar	Admin	Yes	100%	A	Enrollment, teacher qual.
PPM + PAM	Admin	Yes	100%	A	Land area, livestock value
CAGED	Admin	Mo→Yr	~98%	B	Job flows, turnover
Censo Agropecuário	Census	Decennial	~All	B	Capital, land use (2006, 2017)
Censo Demográfico	Census	Decennial	All	B	Informality, total emp.
PNAD Contínua	Survey	Quarterly	States only	D	<i>Infeasible — no muni ID</i>

All Grade-A sources available on basedosdados with `muni_id` $\times$ `year` join keys. Grade-B sources useful as baseline controls (decennial) or flow complements (CAGED). PNAD municipality identifiers suppressed in public microdata — not usable for full panel.

Immediate action: extract RAIS education/age variables (script 22 column expansion, zero data access cost), add INEP + PPM/PAM via BigQuery. Awaiting advisor decision on priorities.

Selected Regression Tables

Agg Split: BNDES Extensive, Coalition, MxJ+MxT — Top-Quartile Municipalities

BNDES sector · Unweighted · Cycle-specific · Pooled-count

	Dep. var: Share of firms receiving a BNDES loan _{jmt}					
	M	G	P	M+G	M+P	M+G+P
$\overline{FA}_{coal.}^{mayor}$	-0.003 (0.003)			-0.001 (0.003)	0.000 (0.003)	-0.001 (0.003)
$\overline{FA}_{coal.}^{gov}$		0.012** (0.002)		0.013** (0.002)		0.013** (0.002)
$\overline{FA}_{coal.}^{pres}$			-0.016 (0.016)		-0.012 (0.014)	-0.012 (0.014)
Exposure Control	Yes	Yes	Yes	Yes	Yes	Yes
F-statistic	1.2	23.5	1.0	16.1	0.4	13.4
Observations	63,205	54,330	54,330	53,311	53,311	53,311
R ²	0.665	0.697	0.697	0.708	0.708	0.708

Muni × sector + muni × year FE. Coalition alignment. Cycle-specific baseline. Pooled-count exposure. Owner-count aggregation. Unweighted regressions. Sample: top-quartile municipalities by mean RAIS employment (2002–2017). Exposure controls interacted with year. SEs clustered by muni + BNDES sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Firm: log(Employment) — Coalition · Cycle-specific · Unweighted · All Firms

Pooled-count exposure · Firm + muni×year FE

	Dep. var: log(Employment _{fmt})					
	M	G	P	M+G	M+P	M+G+P
$FA_{coal.}^{mayor}$	0.027*** (0.002)			0.025*** (0.002)	0.026*** (0.002)	0.024*** (0.002)
$FA_{coal.}^{gov}$		0.021*** (0.002)		0.017*** (0.002)		0.016*** (0.002)
$FA_{coal.}^{pres}$			0.015*** (0.003)		0.010*** (0.003)	0.006** (0.003)
<i>F</i> -statistic	188.5	87.6	27.2	103.8	94.0	70.8
Observations	39,534,589	39,534,589	39,534,589	39,534,589	39,534,589	39,534,589
<i>R</i> ²	0.776	0.776	0.776	0.776	0.776	0.776

Firm + muni × year FE. Coalition alignment. Cycle-specific baseline. Pooled-count exposure. Sample: employment > 0. SEs clustered by firm + muni in parentheses. ****p* < 0.01, ***p* < 0.05, **p* < 0.10.

Firm Split: BNDES Extensive — Top-Quartile Municipalities

Cycle-specific · Unweighted · Coalition · Pooled-count

	Dep. var: $1(\text{BNDES}_{fmt} > 0)$					
	M	G	P	M+G	M+P	M+G+P
$FA_{\text{coal.}}^{\text{mayor}}$	0.002*** (0.000)			0.001*** (0.000)	0.002*** (0.000)	0.001*** (0.000)
$FA_{\text{coal.}}^{\text{gov}}$		0.003*** (0.000)		0.003*** (0.000)		0.003*** (0.000)
$FA_{\text{coal.}}^{\text{pres}}$			0.000 (0.000)		0.000 (0.000)	-0.000 (0.000)
<i>F</i> -statistic	21.6	63.9	1.3	37.1	11.3	24.9
Observations	37,924,455	37,924,455	37,924,455	37,924,455	37,924,455	37,924,455
R^2	0.287	0.287	0.287	0.287	0.287	0.287

Firm + muni × year FE. Coalition alignment. Cycle-specific baseline. Pooled-count exposure. Sample: top-quartile municipalities by mean RAIS employment (2002–2017). SEs clustered by firm + muni in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Firm Split: BNDES Extensive — Bottom Three Quartiles

Cycle-specific · Unweighted · Coalition · Pooled-count

	Dep. var: $1(\text{BNDES}_{fmt} > 0)$					
	M	G	P	M+G	M+P	M+G+P
$FA^{\text{mayor}}_{\text{coal.}}$	0.003*** (0.001)			0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)
$FA^{\text{gov}}_{\text{coal.}}$		0.005*** (0.001)		0.005*** (0.001)		0.005*** (0.001)
$FA^{\text{pres}}_{\text{coal.}}$			0.000 (0.001)		-0.000 (0.001)	-0.002* (0.001)
F-statistic	33.3	61.1	0.1	40.6	16.8	28.8
Observations	5,197,189	5,197,189	5,197,189	5,197,189	5,197,189	5,197,189
R^2	0.319	0.319	0.319	0.319	0.319	0.319

Firm + muni × year FE. Coalition alignment. Cycle-specific baseline. Pooled-count exposure. Sample: bottom three quartiles by mean RAIS employment (2002–2017). SEs clustered by firm + muni in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Agg Full: Emp Share, Party, MxJ+MxT — Full Sample

BNDES sector · Equal-firm · Unweighted · Cycle-specific · Ctrl

	Dep. var: Emp share _{jmt}					
	M	G	P	M+G	M+P	M+G+P
$\overline{FA}_{\text{party}}^{\text{mayor}}$	0.020 (0.015)			0.021 (0.015)	0.019 (0.015)	0.019 (0.014)
$\overline{FA}_{\text{party}}^{\text{gov}}$		-0.013* (0.005)		-0.014* (0.006)		0.001 (0.006)
$\overline{FA}_{\text{party}}^{\text{pres}}$			-0.067** (0.015)		-0.065** (0.016)	-0.065** (0.017)
Exposure Control	Yes	Yes	Yes	Yes	Yes	Yes
F-statistic	1.7	8.5	18.7	3.6	12.8	15.8
Observations	253,086	253,086	253,086	253,086	253,086	253,086
R ²	0.950	0.949	0.950	0.950	0.950	0.950

Muni × sector + muni × year FE. Party alignment. Cycle-specific baseline. Pooled-count exposure. Equal-firm aggregation. Unweighted regressions. Exposure controls interacted with year. SEs clustered by muni + BNDES sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Agg Full: Emp Share, Party, MxJ+JxT — Full Sample

BNDES sector · Equal-firm · Unweighted · Cycle-specific · Ctrl

	Dep. var: Emp share _{jmt}					
	M	G	P	M+G	M+P	M+G+P
$\overline{FA}_{\text{party}}^{\text{mayor}}$	-0.006 (0.003)			-0.006 (0.004)	-0.007 (0.004)	-0.007 (0.004)
$\overline{FA}_{\text{party}}^{\text{gov}}$		-0.001 (0.008)		0.000 (0.008)		0.007 (0.012)
$\overline{FA}_{\text{party}}^{\text{pres}}$			-0.036** (0.010)		-0.036** (0.011)	-0.038* (0.014)
Exposure Control	Yes	Yes	Yes	Yes	Yes	Yes
F-statistic	3.5	0.0	12.4	2.6	6.5	20.0
Observations	255,212	255,212	255,212	255,212	255,212	255,212
R ²	0.953	0.953	0.953	0.953	0.953	0.953

Muni × sector + sector × year FE. Party alignment. Cycle-specific baseline. Pooled-count exposure. Equal-firm aggregation. Unweighted regressions. Exposure controls interacted with year. SEs clustered by muni + BNDES sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Agg Full: log(Emp), Party, MxJ+MxT — Full Sample

BNDES sector · Equal-firm · Unweighted · Cycle-specific · Ctrl

	Dep. var: log(Emp _{jmt})					
	M	G	P	M+G	M+P	M+G+P
$\overline{FA}_{\text{party}}^{\text{mayor}}$	0.369 (0.292)			0.379 (0.295)	0.358 (0.284)	0.357 (0.284)
$\overline{FA}_{\text{party}}^{\text{gov}}$		-0.174 (0.108)		-0.203 (0.115)		0.023 (0.062)
$\overline{FA}_{\text{party}}^{\text{pres}}$			-1.010** (0.311)		-0.965** (0.285)	-0.972** (0.287)
Exposure Control	Yes	Yes	Yes	Yes	Yes	Yes
F-statistic	1.6	2.6	10.6	1.6	12.6	8.6
Observations	253,086	253,086	253,086	253,086	253,086	253,086
R ²	0.948	0.947	0.948	0.948	0.948	0.948

Muni × sector + muni × year FE. Party alignment. Cycle-specific baseline. Pooled-count exposure. Equal-firm aggregation. Unweighted regressions. Exposure controls interacted with year. SEs clustered by muni + BNDES sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Agg Full: $\log(\text{Emp})$, Party, $\text{MxJ} + \text{JxT}$ — Full Sample

BNDES sector · Equal-firm · Unweighted · Cycle-specific · Ctrl

	Dep. var: $\log(\text{Emp}_{jmt})$					
	M	G	P	M+G	M+P	M+G+P
$\overline{FA}_{\text{party}}^{\text{mayor}}$	-0.011 (0.039)			-0.003 (0.046)	-0.022 (0.030)	-0.024 (0.028)
$\overline{FA}_{\text{party}}^{\text{gov}}$		-0.167 (0.135)		-0.164 (0.135)		0.041 (0.047)
$\overline{FA}_{\text{party}}^{\text{pres}}$			-1.080 (0.648)		-1.075 (0.653)	-1.086 (0.665)
Exposure Control	Yes	Yes	Yes	Yes	Yes	Yes
F-statistic	0.1	1.5	2.8	8.0	13.7	9.1
Observations	255,212	255,212	255,212	255,212	255,212	255,212
R ²	0.943	0.943	0.943	0.943	0.943	0.943

Muni $\times$ sector + sector $\times$ year FE. Party alignment. Cycle-specific baseline. Pooled-count exposure. Equal-firm aggregation. Unweighted regressions. Exposure controls interacted with year. SEs clustered by muni + BNDES sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Agg: Employment Share, Party, MxJ+MxT — Top-Quartile Municipalities

BNDES sector · Equal-firm aggregation · Unweighted · Cycle-specific · Pooled-count

	Dep. var: Emp share _{jmt}					
	M	G	P	M+G	M+P	M+G+P
$\overline{FA}_{\text{party}}^{\text{mayor}}$	0.044 (0.039)			0.047 (0.038)	0.047 (0.037)	0.048 (0.038)
$\overline{FA}_{\text{party}}^{\text{gov}}$		-0.020 (0.034)		-0.018 (0.035)		-0.011 (0.036)
$\overline{FA}_{\text{party}}^{\text{pres}}$			-0.134*** (0.021)		-0.138*** (0.023)	-0.138*** (0.022)
Exposure Control	Yes	Yes	Yes	Yes	Yes	Yes
F-statistic	1.3	0.3	41.7	0.8	18.5	13.7
Observations	71,761	71,761	71,761	71,761	71,761	71,761
R ²	0.930	0.930	0.930	0.930	0.930	0.930

Muni × sector + muni × year FE. Party alignment. Cycle-specific baseline. Pooled-count exposure. Equal-firm aggregation. Unweighted regressions. Sample: top-quartile municipalities by mean RAIS employment (2002–2017). Exposure controls interacted with year. SEs clustered by muni + BNDES sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.